Practice VI.

1. 求模23的二次剩余和非二次剩余.

Z_{23}^* 中, $1^2, 2^2, \cdots \left(\frac{23-1}{2}\right)^2$ 是 Z_{23}^* 中的全部二次剩余.

模23的二次剩余: $\{1, 4, 9, 16, 2, 13, 3, 18, 12, 8, 6\}$

模23的非二次剩余: $\{5, 7, 10, 11, 14, 15, 17, 19, 20, 21, 22\}$.

2. 指出以下同余式解的个数:

① $x^2 \equiv 2 \pmod{31}$

$$\left(\frac{2}{31}\right) = \begin{cases} 1 & P \equiv \pm 1 \pmod 8 \\ -1 & P \equiv \pm 3 \pmod 8 \end{cases}$$

$P = 31 \equiv -1 \pmod 8$ 所以 $\left(\frac{2}{31}\right) = 1$

又 $P \equiv 3 \pmod 4$, 则 $\pm n^{\frac{1}{4}(p+1)}$ 是该同余式的2个解.

② $x^2 \equiv 3 \pmod{31}$

$\left(\frac{3}{31}\right) \equiv 3^{\frac{31-1}{2}} \equiv 3^{15} \pmod{31}$ 判断 $3^{15} \bmod 31$ $3^2 \equiv 9 \pmod{31}$ $3^4 \equiv 19 \pmod{31}$ $3^8 \equiv 20 \pmod{31}$

$3^{15} \equiv 20 \cdot 19 \cdot 9 \cdot 3 \pmod{31}$ $3^{15} \equiv -1 \pmod{31}$

所以 $\left(\frac{3}{31}\right)$ 不是31的二次剩余 所以该同余式无解.

③ $x^2 \equiv 19 \pmod{30}$

由中国剩余定理:
$$\begin{cases} x^2 \equiv 19 \pmod 2 \Rightarrow x^2 \equiv 1 \pmod 2 \Rightarrow x \equiv 1 \pmod 2 \\ x^2 \equiv 19 \pmod 3 \Rightarrow x^2 \equiv 1 \pmod 3 \Rightarrow x \equiv 1, 2 \pmod 3 \\ x^2 \equiv 19 \pmod 5 \Rightarrow x^2 \equiv 4 \pmod 5 \Rightarrow x \equiv 2, 3 \pmod 5 \end{cases}$$

共 $1 \times 2 \times 2 = 4$ 个解.

3. 计算 $\left(\frac{7}{19}\right)$ 7 2×7 3×7 4×7 5×7 6×7 7×7 8×7 9×7 mod 19 $\frac{19}{2} = 9.5$

有4个大于 $\frac{p}{2}$ 故 $\left(\frac{7}{19}\right) = (-1)^4 = 1$

4. 即判断 $x^2 \equiv -2 \pmod p$ 的有解情况与无解情况

$\left(\frac{-2}{p}\right)=\left(\frac{2}{p}\right)\left(\frac{-1}{p}\right)$

$\left(\frac{-1}{p}\right)=\begin{cases}1 & p\equiv 1 \pmod 4\\ -1 & p\equiv 3 \pmod 4\end{cases}$ $\left(\frac{2}{p}\right)=\begin{cases}1 & p\equiv \pm 1 \pmod 8\\ -1 & p\equiv \pm 3 \pmod 8\end{cases}$

$p\equiv 1,3,5,7 \pmod 8$

$p\equiv 1 \pmod 8$ 时 $\left(\frac{-1}{p}\right)=1$ $\left(\frac{2}{p}\right)=1$ $\Rightarrow \left(\frac{-2}{p}\right)=1$

$p\equiv 3 \pmod 8$ $\left(\frac{-1}{p}\right)=-1$ $\left(\frac{2}{p}\right)=-1$ $\Rightarrow \left(\frac{-2}{p}\right)=1$

$p\equiv 5 \pmod 8$ $\left(\frac{-1}{p}\right)=1$ $\left(\frac{2}{p}\right)=-1$ $\Rightarrow \left(\frac{-2}{p}\right)=-1$

$p\equiv 7 \pmod 8$ $\left(\frac{-1}{p}\right)=-1$ $\left(\frac{2}{p}\right)=1$ $\Rightarrow \left(\frac{-2}{p}\right)=-1$

所以 $p\equiv 1,3 \pmod 8$ 是 -2 模 p 的二次剩余 $p\equiv 5,7 \pmod p$ 是非 -2 模 p 的二次剩余.

5. ① $\left(\frac{17}{99}\right)=(-1)^{8\cdot\frac{99-1}{2}}\cdot\left(\frac{99}{17}\right)$ $\left(\frac{99}{17}\right)=\left(\frac{14}{17}\right)=\left(\frac{2}{17}\right)\left(\frac{7}{17}\right)=\left(\frac{7}{17}\right)=\left(\frac{17}{7}\right)=\left(\frac{3}{7}\right)=-\left(\frac{7}{3}\right)=-1$

② $\left(\frac{610}{987}\right)=\left(\frac{2}{987}\right)\left(\frac{305}{987}\right)=-\left(\frac{305}{987}\right)=-\left(\frac{72}{305}\right)=-\left(\frac{2}{305}\right)^3\cdot\left(\frac{9}{305}\right)=-\left(\frac{9}{305}\right)=-\left(\frac{8}{9}\right)=-\left(\frac{2}{9}\right)^3=-1$

6. $p-1=2^n$ 即证 $3^{p-1}\equiv 1 \pmod p$ 只需证明 $3^{2^n}\equiv 1 \pmod p$

下证 n 必须是偶数，若 n 是奇数.

$2^n+1 \equiv (-1)^n+1 \equiv 0 \pmod 3$ 这说明 $3\mid 2^n+1$ 即 2^n+1 一定有因子 3，但与 2^n+1 是素数矛盾

所以 n 为偶 $2^n+1\equiv 5 \pmod{12}$ 即 $p\equiv 5 \pmod{12}$ 所以 $\left(\frac{3}{p}\right)=-1$

$\left(\frac{3}{p}\right)\equiv 3^{\frac{p-1}{2}} \pmod p$ $3^{\frac{p-1}{2}}\equiv -1 \pmod p$ $3^{2^{n-1}}\equiv -1 \pmod p$

由费马小定理：$3^{2^n}\equiv 1 \pmod p$ 但 $3^{2^{n-1}}\equiv -1 \pmod p$ 所以 3 的阶所只能是 2^n 即 $p-1$